

1	Course Name:
	Electronics
2	Course Code:
	IBB22503
3	Year and Semester Offered:
	Year 2 Semester 1
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IBB12704 Electrical Circuit Analysis
6	Course Synopsis:
	This course will enable students to comprehend the basic operation and various applications of semiconductor devices appropriated to analog electronics system. Students will be introduced to the device's operating principles and their main application in DC and AC circuits. At the same time, students will be exposed to the current measuring instruments used to investigate and analyse the behaviour and characteristics of such devices.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain the operating principles of different semiconductor electronics devices. 2. Demonstrate the acquired knowledge to analyze the operation and applications of the semiconductor electronics devices through calculations, practical or given problems. 3. Construct simple functional analog electronics system by employing semiconductor electronics devices
8	Content Outline:
	Topic
	1. Solid State Principles 1.1 Atomic Theory 1.2 Properties of Conductor Semiconductor and Insulator 1.3 Electrical circuit configurations and circuit elements 1.4 Doping & PN-Junction 1.5 Biasing 2. PN-Junction Diode & Applications 2.1 Construction, Operating Principles & Diode Model 2.2 Characteristic Curve and Data Sheet 2.3 DC Power Supply 2.4 Applications - Rectifier, Clipper and Clamper 3. Special Purpose Diodes 3.1 Zener Diode - Operating Principles & Applications 3.2 LED - Operating Principles & Applications

	4. Bipolar Junction Transistor (BJT)
	4.1 Construction, Operating Principles, Characteristic Curve
	4.2 Data sheets, Electrical characteristics and parameters
	4.3 DC load line and DC biased circuits
	4.4 Applications - Amplifier, Switching Device
	5. Field Effect Transistor (FET)
	5.1 JFET characteristics and parameters
	5.2 MOSFET characteristics and parameters
	5.3 FET's applications
	6. Operational Amplifier (Op-Amp)
	6.1 Introduction to Op-Amp and OpAmp's Model
	6.2 Operating principles – Gain, Input/Output, Supply voltage
	6.3 Op-Amp's Data sheets – Characteristics & Parameters
	6.4 Application – Inverting, Non-inverting amplifier, Voltage follower, Summing, Integrator and Differentiator